

SINGAPORE PAYA LEBAR, Singapore

WMO: 486940

Lat: 1.360N

Lon: 103.910E

Elev: 20

StdP: 101.09

Time Zone: 8.00 (E08)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	23.9	19.9	14.7	28.9	20.8	15.5	28.5	9.1	29.6	8.4	29.6	2.1	10	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.0	34.2	26.3	33.8	26.3	33.1	26.2	28.1	31.1	27.7	30.7	27.4	30.5	3.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.3	27.0	22.7	30.1	26.2	21.7	29.3	89.8	32.1	88.2	31.2	86.9	30.8	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	5.7	DB	22.6	35.3	0.7	0.6	22.1	35.7	21.7	36.1	21.3	36.4	20.8	36.9	(4)
(5)			WB	21.0	29.0	0.8	1.0	20.5	29.7	20.0	30.3	19.6	30.9	19.1	31.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	27.9	28.4	29.0	29.4	29.6	29.3	28.9	28.8	28.9	29.0	28.6	28.2	(6)
(7)		DBStd	1.29	1.33	1.05	1.20	1.18	1.16	1.30	1.28	1.25	1.17	1.18	1.03	1.22	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6876	556	516	588	583	608	578	584	583	566	589	559	564	(10)
(11)		CDD18.3	3834	298	283	330	333	350	328	326	324	316	331	309	306	(11)
(12)		CDH23.3	43464	2954	2934	3638	3871	4346	4063	3869	3839	3722	3839	3287	3102	(12)
(13)		CDH26.7	16713	908	1005	1392	1606	1939	1748	1537	1496	1464	1530	1122	965	(13)
(14)	Wind	WSAvg	2.5	3.4	3.6	2.7	1.6	1.9	2.2	2.6	2.7	2.4	1.9	1.7	2.7	(14)
(15)	Precipitation	PrecAvg	2153	180	175	161	150	156	146	144	141	159	176	234	324	(15)
(16)		PrecMax	2954	483	470	361	346	265	255	255	258	344	431	464	670	(16)
(17)		PrecMin	1562	22	6	19	17	77	59	29	52	47	39	67	125	(17)
(18)		PrecStd	428	119	128	91	83	44	53	53	54	68	107	100	155	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.9	34.2	35.1	34.9	34.2	33.2	33.2	33.8	34.2	34.0	33.9	(19)
(20)			MCWB	26.1	26.4	26.4	26.7	26.6	25.8	26.0	25.7	25.8	25.6	26.4	26.2	(20)
(21)		2%	DB	32.2	33.0	33.8	34.1	33.9	33.2	32.8	32.8	33.0	33.2	33.0	32.8	(21)
(22)			MCWB	25.8	26.1	26.3	26.8	26.6	26.1	26.0	25.8	25.7	25.8	26.2	26.1	(22)
(23)		5%	DB	31.8	32.1	32.9	33.1	33.0	32.8	32.0	32.0	32.1	32.8	32.1	31.9	(23)
(24)			MCWB	25.8	25.7	26.1	26.6	26.5	26.1	25.9	25.8	25.7	25.8	26.0	26.0	(24)
(25)		10%	DB	30.9	31.2	32.0	32.2	32.1	31.9	31.1	31.1	31.2	31.9	31.2	30.9	(25)
(26)			MCWB	25.7	25.6	25.9	26.5	26.6	26.1	26.1	25.9	25.9	25.8	26.2	26.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.4	27.9	28.5	28.0	28.0	27.9	27.5	27.9	28.0	27.8	(27)	
(28)			MADB	31.0	31.7	31.6	32.1	31.6	30.5	30.2	30.4	30.2	30.8	30.8	30.5	(28)
(29)		2%	WB	26.8	27.1	27.2	27.8	27.8	27.4	27.2	27.2	26.9	27.2	27.2	27.1	(29)
(30)			MADB	30.1	30.6	31.0	31.3	30.9	30.2	29.9	29.9	30.1	30.3	30.5	30.3	(30)
(31)		5%	WB	26.5	26.5	26.8	27.3	27.4	27.1	26.8	26.7	26.7	26.7	26.7	26.6	(31)
(32)			MADB	29.6	30.0	30.4	30.8	30.6	30.1	29.7	29.6	29.8	29.9	30.0	29.8	(32)
(33)		10%	WB	26.1	26.1	26.5	26.9	27.1	26.7	26.6	26.5	26.4	26.5	26.4	26.3	(33)
(34)			MADB	29.2	29.5	30.1	30.4	30.3	29.8	29.6	29.5	29.6	29.7	29.6	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.5	6.8	6.8	6.0	5.4	5.2	5.2	5.7	6.2	6.3	5.8	(35)
(36)			MCDBR	6.9	7.3	7.6	7.5	6.6	6.1	5.6	5.7	6.2	7.0	7.1	7.1	(36)
(37)			MCWBR	2.6	2.6	2.7	2.7	2.6	2.5	2.5	2.4	2.6	2.6	2.8	2.7	(37)
(38)		5% WB	MADB	6.3	6.6	7.0	6.9	6.0	5.5	5.2	5.4	5.8	6.2	6.5	6.3	(38)
(39)			MCWBR	2.8	2.9	3.0	3.0	2.8	2.6	2.7	2.7	2.8	2.7	2.8	2.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.470	0.482	0.492	0.489	0.475	0.486	0.478	0.521	0.552	0.579	0.484	0.470	(40)	
(41)		taud	2.297	2.244	2.226	2.234	2.282	2.253	2.283	2.141	2.050	1.955	2.254	2.304	(41)	
(42)		Ebn at Noon	861	856	843	826	815	793	805	789	781	765	842	854	(42)	
(43)		Edh at Noon	137	146	149	144	133	134	131	156	175	193	142	134	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.54	5.26	5.24	5.07	4.69	4.45	4.42	4.49	4.48	4.48	4.23	4.13	(44)	
(45)		RadStd	0.38	0.47	0.46	0.32	0.22	0.32	0.34	0.26	0.31	0.33	0.26	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.57	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.37	+0.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page